

Bluestock Mutual Fund Analytics Capstone

Final Project Report

End-to-end mutual fund analytics pipeline covering ingestion, cleaning, database design, exploratory analysis, performance and risk analytics, advanced investor analytics, recommendations, and Power BI reporting.

1. Project Objectives

- Build a reusable ETL workflow for multiple mutual fund datasets.
- Clean, validate and standardize data while preserving meaningful missing observations.
- Create a relational SQLite database with primary and foreign-key relationships.
- Calculate fund performance and risk metrics using transparent, reproducible methods.
- Analyze investor retention through cohort analysis.
- Develop a rule-based mutual fund recommendation model.
- Present insights through an interactive four-page Power BI dashboard.

2. Data & ETL Pipeline

- The raw data contains fund master information, historical NAV, portfolio holdings, scheme performance, benchmark indices, AUM by fund house, SIP inflows, category inflows, folio counts and investor transactions.
- The ETL pipeline dynamically resolves the project root, reads CSV files from data/raw, removes completely empty rows and columns, removes exact duplicates, standardizes column names and trims text fields, then saves cleaned datasets to data/processed.
- Validation includes missing-value checks, duplicate checks, numeric checks, date validation and AMFI-code consistency.

3. Database Design

- SQLite is used to store the cleaned analytical datasets. The schema includes fund_master, nav_history, aum_by_fund_house, monthly_sip_inflows, category_inflows, industry_folio_count, scheme_performance, portfolio_holdings, investor_transactions and benchmark_indices.
- Foreign-key relationships connect AMFI codes in dependent tables to fund_master, supporting relational integrity.

4. Exploratory Data Analysis

- EDA examines fund-house AUM, SIP inflow trends, category-level inflows, scheme 3-year performance, NAV trends and correlations among major performance variables.
- The analysis supports comparison of funds, categories and market trends before deeper risk analytics.

5. Performance & Risk Analytics

- CAGR is calculated using actual elapsed years rather than assuming a fixed 252-day holding period.
- Annualized volatility uses daily return standard deviation scaled by the square root of 252.
- Maximum drawdown measures the largest peak-to-trough decline.
- Historical 95% daily VaR estimates the one-day loss threshold from the empirical return distribution, using a positive loss convention.
- Beta is calculated against an aligned benchmark return series. Sharpe ratio uses a zero risk-free-rate assumption because a risk-free series was not included in the supplied data.

6. Advanced Analytics

- Cohort analysis groups investors by their first transaction month and measures subsequent activity by months since cohort formation.
- The recommendation model uses rank-based weighted scoring: 30% 3-year return, 25% Sharpe ratio, 20% volatility, 15% drawdown and 10% expense ratio. Higher return and Sharpe are preferred; lower volatility, drawdown magnitude and expense ratio are preferred.

7. Power BI Dashboard

- The dashboard has four pages: Executive Overview; Fund Performance & Risk; Investor & SIP Analytics; and Advanced Analytics.
- Interactive slicers enable users to filter the analytical views by relevant fund-house, category and investor dimensions.

8. Deliverables

- README.md — project documentation and structure.
- Final_Report.pdf — project methodology, analytics and dashboard summary.
- Presentation.pptx — concise project presentation.
- bluestock_mf.pbix — interactive Power BI dashboard.
- Python notebooks, scripts and SQL schema/query files supporting the analysis.

9. Conclusion

- The project provides a reproducible workflow from raw mutual fund data to validated analytics and an interactive business dashboard. The combination of performance, risk, investor behavior and recommendation analytics supports practical mutual fund research and decision-making.